



COURSE DESCRIPTION		
Course Code	26SE101	Course Credit: 4
Course Name	Distributed Systems and Advanced Computing	
Course Delivery Mode	TiC	
L-T-P-H	4-0-0-0	
Pre - requisite	Nil	
Year of Introduction and Syllabus Version	2026 V1.0	

COURSE OBJECTIVES		Cognitive Level	Knowledge Category
1	To understand the fundamentals, architectures, and design goals of distributed systems.	L2	K2
2	To analyze grid computing infrastructure, resource management, and scheduling techniques.	L3	K3
3	To apply cluster computing concepts for parallel processing and high - performance systems.	L3	K3
4	To evaluate cloud computing models, services, and infrastructure challenges.	L4	K4
5	To develop skills in deploying and managing cloud platforms such as OpenStack	L4	K4

COURSE OUTCOMES (COs)	
After the successful completion of the course students will be able to:	
CO1	Explain distributed system architectures, consistency models, and replication techniques.
CO2	Analyze grid computing components, resource management, and scheduling mechanisms.
CO3	Apply cluster computing techniques for parallel and fault - tolerant systems.
CO4	Evaluate cloud computing models, services, and associated risks.
CO5	Implement and manage cloud platforms such as OpenStack for real - world applications.

CO - PO MAPPING						
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	2	1	-	1
CO2	3	2	3	1	1	1
CO3	3	2	3	2	2	2
CO4	2	2	3	2	2	2
CO5	2	2	3	2	3	2
Correlation levels: 1 - Low; 2 - Medium; 3 - High; No Correlation - " - "						





COURSE IMPLEMENTATION CLOCK (IN HOURS)	
Zeroth Module	1 Hr.
Course content delivery hours	48 Hrs.
Internal Test (2 Assessment Tests of one and a half hour duration)	3 Hrs.

COURSE CONTENTS

Zeroth Module: General Introduction to the Course			
- Course introduction with a thorough briefing of Course Objectives and Course Outcomes - An overview of the POs that are mapped to the course		No ESE	1 Hr.
Module - I		COs	Hrs.
Module Title	Introduction to Distributed Systems		
Distributed versus decentralized systems - Design goals of distributed systems - Classifications of distributed systems - Architectural styles - Consistency and replication - Data - centric consistency models - Client - centric consistency models - Replica management		1	10
Module - II		COs	Hrs.
Module Title	Grid Computing		
Introduction - Characteristics of grid computing - Grid computing infrastructures - Virtual organizations and the grid - Grid architecture - Implementing grid architecture - CPU scavenging and virtual supercomputers - Data - intensive grid service models - The grid community - Open Grid Services Architecture (OGSA) - Grid resource management and brokering - Resource management and job scheduling - Grid resource monitoring with CGSP - Service accounting and economy model - Resource brokering with Gridbus		2	10
Module - III		COs	Hrs.
Module Title	Cluster Computing		
Introduction - Clustering for massive parallelism - Design objectives of computer cluster - Fundamental cluster design issues - Computer clusters and MPP architectures - Cluster organization and resource sharing - Node architectures and MPP packaging - Cluster system interconnects - GPU clusters for massive parallelism - Design principles of computer clusters - Single - system image features - High availability through redundancy - Fault - tolerant cluster configurations - Cluster job and resource management		3	10
Module - IV		COs	Hrs.
Module Title	Cloud Computing		
Introduction to cloud computing - Roots of cloud computing - Layers and types of clouds - Infrastructure as a Service - Platform as a Service - Software as a Service - Deployment models - Desired features of a cloud - Cloud infrastructure management - Challenges and risks in the cloud computing model.		4	9





Module – V		COs	Hrs.
Module Title	Cloud computing platforms		
OpenStack - Architecture - Components - OpenStack deployment with DevStack - Creation of flavours - Concepts of Neutron networking - Images management - User managing with Keystone - Concepts of object store - OpenStack compute - Nova OpenStack Neutron		5	9

TEXT BOOKS

- 1 Maarten van Steen and Andrew S. Tanenbaum, *Distributed Systems*, 4th Edition, Pearson Education, 2023 ISBN: 978 - 90 - 815406 - 3 - 6.
- 2 Kai Hwang, Jack Dongarra , Geoffrey C. Fox *Distributed and Cloud Computing From Parallel Processing to the Internet of Things*, 1st Edition, Elsevier, 2013, ISBN: 978 - 0128002049.
- 3 Kevin Jackson, Cody Bunch, *OpenStack Cloud Computing Cookbook*, 2nd Edition, Packt Publishing, 2016, ISBN - 13: 978 - 1782167587.

REFERENCE BOOKS/ WEB REFERENCES

- 1 Tanenbaum, A.S., and van Steen, M. *Distributed Systems: Principles and Paradigms*, 2nd Edition, Prentice, Hall, 2007, ISBN: 978,0,13,239227.
- 2 'Dan C. Marinescu, *Cloud Computing: Theory and Practice*, Elsevier, 2013 ISBN - 13: 978 - 0124046276.
- 3 Thomas Erl, Ricardo Puttini, Zaigham Mahmood, *Cloud Computing: Concepts, Technology & Architecture*, 2nd Edition, Pearson Publications, 2013 ISBN: 9780133387520 .
- 4 Michael Solberg, *OpenStack for Architects*, 2nd Edition , Packt Publishing, 2018 ISBN - 13: 978 - 788395410.
- 5 <https://www.openstack.org/>

MODES OF EVALUATION		SCORE WEIGHTAGE / SPLIT MARKS	
Continuous Internal Evaluation (CIE)		40	
Internal Test - I		12.5	
Internal Test - II		12.5	
Taskworks		15	
Assignment	50%		
Seminar / Quiz	50%		
End Semester Examination (ESE)		60	
Total		100	
End Semester Examination Pattern			
The question paper consists of two parts: Part A: 15 marks Part B: 45 marks Part A contains a total of 5 questions, 1 question each from each module, having 3 marks. Part B contains a total of 10 questions, 2 questions from each module of which students should answer any one. Each question in part B carries 9 marks and can have a maximum of 2 sub - divisions.			





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SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS)
M.Tech. SYLLABUS 2026

APPROVAL SUMMARY	
	Number and Date
Recommended by BoS	SCE/BoS/CS/M.Tech./SE/013/25- 26, 22-05-26
Approved by Academic Council	AC/RES/019/25-26, 15-06-2026

APPROVED BY ACADEMIC COUNCIL





COURSE DESCRIPTION		
Course Code	26SE102	Course Credit: 3
Course Name	Advanced Data Structures & Algorithm Design	
Course Delivery Mode	Tic/MOOC	
L-T-P-H	3-0-0-0	
Pre - requisite	Nil	
Year of Introduction and Syllabus Version	2026 V1.0	

COURSE OBJECTIVES		Cognitive Level	Knowledge Category
1	To introduce fundamental concepts of algorithm design, complexity analysis, and asymptotic behaviour.	L2	K1
2	To develop understanding of core data structures such as stacks, queues, trees, and their applications.	L3	K2
3	To enable the application of advanced data structures and algorithm design paradigms like Divide and Conquer and Greedy methods.	L3	K3
4	To analyse and evaluate graph algorithms, optimization techniques, and advanced sorting and selection methods.	L4	K4

COURSE OUTCOMES (COs)	
After the successful completion of the course students will be able to:	
CO1	Explain algorithmic concepts, complexity measures, and asymptotic analysis techniques.
CO2	Apply fundamental data structures such as stacks, queues, and binary search trees to solve problems.
CO3	Design algorithms using advanced paradigms like Divide and Conquer and implement balanced trees and heaps.
CO4	Analyse graph algorithms and advanced techniques for solving optimization and real - world computational problems.

CO - PO MAPPING						
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	2	-	-	1
CO2	3	1	3	-	1	2
CO3	3	2	3	-	2	2
CO4	3	2	3	1	2	2
Correlation levels: 1 - Low; 2 - Medium; 3 - High; No Correlation - " - "						